

HW 6.4.2: The Dot Product

1. Given $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{v} = \langle -1, 8 \rangle$ find the following.

a. $\mathbf{u} \cdot \mathbf{v} =$

b. The angle between $\mathbf{u}$ and $\mathbf{v}$.

2. Given $\mathbf{u} = \langle 1, 6 \rangle$ and $\mathbf{v} = \langle 5, 3 \rangle$ find the following.

a. $\mathbf{u} \cdot \mathbf{v} =$

b. The angle between $\mathbf{u}$ and $\mathbf{v}$.

3. Given $\mathbf{u} = \langle -5, 3 \rangle$ and $\mathbf{v} = \langle 4, 6 \rangle$ find the following.

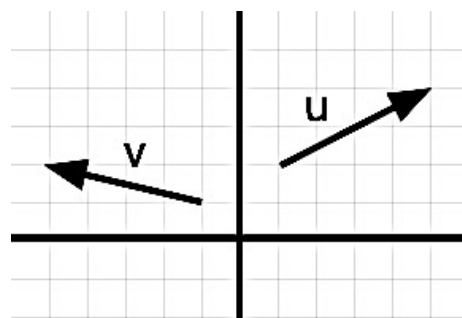
a. $\mathbf{u} \cdot \mathbf{v} =$

b. The angle between $\mathbf{u}$ and $\mathbf{v}$.

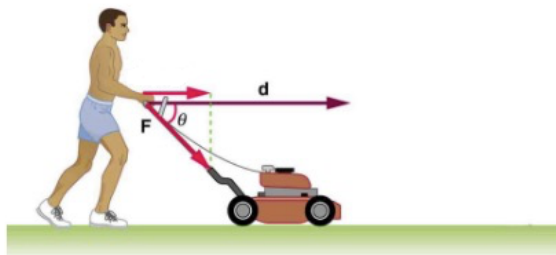
4. Given $\mathbf{u}$ and $\mathbf{v}$ on the right find the following.

a. $\mathbf{u} \cdot \mathbf{v} =$

b. The angle between $\mathbf{u}$ and $\mathbf{v}$.



5. How much work is done on the lawn mower in Figure below if he exerts a constant force of 75.0 N at an angle 35° below the horizontal and pushes the mower 25.0 m on level ground?



6. How much work must be done to lift the 10 kg block up 5.0 meters? (Remember that $F = ma$.)

7. A force of 20 Newtons is applied to an object at an angle of -35° with the horizontal. The object is pulled 10 meters at an angle of 25° . How much work is done while moving the object?

8. Compare and contrast the Law of Cosines with the equation you used to determine the angle between two vectors.

Selected Answers:

2.) a.) 23 b.) 49.57°

$$W = \|\vec{F}\| \|\vec{d}\| \cos \theta$$

$$W = 75 \cdot 25 \cdot \cos(-35^\circ)$$

$$W = 1535.91 \text{ joules}$$

OR

5.) $W = \vec{F} \cdot \vec{d}$

$$W = \langle 75 \cos(-35^\circ), 75 \sin(-35^\circ) \rangle \cdot \langle 25 \cos(0), 75 \sin(0) \rangle$$

$$W = \langle 61.436, -61.436 \rangle \cdot \langle 25, 0 \rangle$$

$$W = 61.436 \cdot 25 + -61.436 \cdot 0$$

$$W = 1535.91 \text{ joules}$$

$$W = \vec{F} \cdot \vec{d}$$

$$W = \langle 10(9.8) \cos(90^\circ), 10(9.8) \sin(90^\circ) \rangle \cdot \langle 0, 5 \rangle$$

6.)

$$W = \langle 0, 98 \rangle \cdot \langle 0, 5 \rangle$$

$$W = 490 \text{ joules}$$

$$W = \vec{F} \cdot \vec{d}$$

$$W = \langle 20 \cos(-35^\circ), 20 \sin(-35^\circ) \rangle \cdot \langle 10 \cos(25^\circ), 10 \sin(25^\circ) \rangle$$

$$W = \langle 16.383, -11.472 \rangle \cdot \langle 9.063, 4.226 \rangle$$

$$W = 100 \text{ joules}$$

7.) OR

$$W = \|\vec{F}\| \|\vec{d}\| \cos \theta$$

$$W = 20 \cdot 10 \cdot \cos(60^\circ)$$

$$W = 20 \cdot 10 \cdot \frac{1}{2} = 100 \text{ joules}$$